



**BOTSWANA UNIVERSITY OF AGRICULTURE AND NATURAL RESOURCES**

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**IRRIGATION AND WATER QUALITY**

**LABORATORY REPORT 1**

## **INTRODUCTION**

Soil chemical properties, particularly pH and Electrical Conductivity (EC) are fundamental indicators of soil health and agricultural productivity. These parameters influence the environment in which plant roots develop and determine the availability of essential nutrients required for plant growth.

### **Soil pH**

Soil pH is a measure of the acidity or alkalinity of the soil solution. It is a critical factor in determining nutrient availability and overall soil fertility. At extremely low or high pH levels, certain nutrients become chemically unavailable to plants. Soil pH also influences microbial activity, as most soil microorganisms responsible for organic matter decomposition function best in near neutral pH conditions.

### **Electrical Conductivity (EC)**

Electrical Conductivity (EC) measures the ability of a soil solution to conduct an electrical current and is directly related to the concentration of dissolved salts present in the soil. High EC values indicate elevated salt levels which may lead to salinity stress in plants. Excess salts create osmotic pressure that limits the ability of plant roots to absorb water even when moisture is present in the soil. Monitoring EC is therefore essential in assessing soil salinity and guiding irrigation and soil management practices to prevent soil degradation.

### **AIM**

The aim of this experiment is to determine the soil pH and Electrical Conductivity (EC) using saturated paste extract (SPE) and soil water extract methods (1:1 and 1:5 ratios) for soil samples collected at different depths in order to evaluate the soil's chemical condition and salinity status.

## **OBJECTIVES**

- To measure the soil pH and Electrical Conductivity (EC) of soil samples using saturated paste extract (SPE) and soil water extract methods (1:1 and 1:5 ratios).
- To compare the pH and EC values at different soil depths (0–15 cm and 15–30 cm) in order to observe vertical variations in soil chemical properties.
- To evaluate the effect of different soil water dilution ratios (1:1 and 1:5) on the measured pH and Electrical Conductivity values.

## **MATERIALS**

### **Equipment & Apparatus**

- BIOBASE 210 Benchtop pH/mV Meter
- Conductivity and pH Electrodes
- Analytical Balance
- Beakers (250 mL)
- Graduated Cylinders
- Whatman #2 Filter Paper
- Büchner Funnel
- Vacuum Pump
- Soil Auger

### **Reagents & Samples**

- Soil Samples: Collected from (0–15 cm) and (15–30 cm)
- Distilled Water
- Standard Buffer Solutions (pH 4.0, 7.0, and 10.0)

## METHODOLOGY



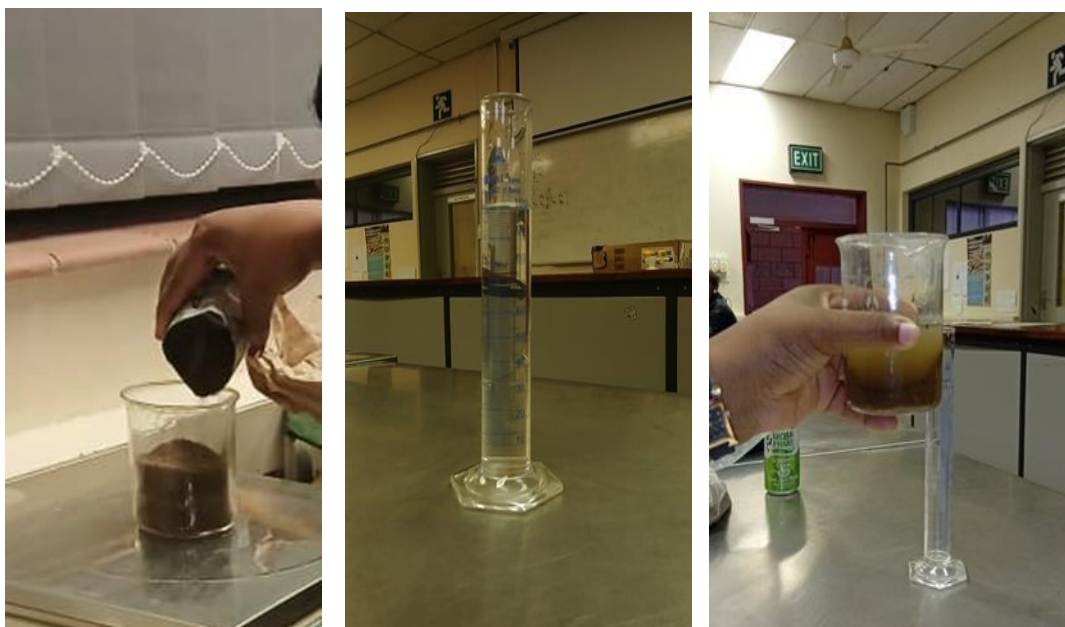
A representative area of the study site was visited using a soil auger a hole was dug to a depth of 15 cm (0 -15 cm), which is usually the most biologically active layer and most affected by fertilization and evaporation. Then after the first 15cm was removed, the hole deepened to the 30 cm mark. The soil collected from the 15–30cm range and this layer is critical for identifying salts that have moved down from the surface. Each depth specific sample was placed in a labeled plastic bag.



The soil samples collected from the field (0–15 cm and 15–30 cm) were spread out in a thin layer of paper and allowed to air-dry. This removes variable water and once dry, the soil was passed through a 2mm sieve. This removes large rocks, roots, and organic debris, leaving only the fine fraction for testing.

For the 1:1 ratio, 100g of this air-dried soil was weighed into a beaker.

For the 1:5 ratio, 15g of this air-dried soil was weighed into a separate beaker.



Once the dry soil was weighed into the beakers, the distilled water was added to begin the extraction of ions. Water Addition to the 100g soil sample, 100 mL of distilled water was added (1:1). And to the 15g soil sample, 75 mL of distilled water was added (1:5). The suspensions were stirred initially to ensure all soil particles were thoroughly wetted. The containers were shaken vigorously by hand for 1 minute. This 1 minute shaking process was repeated 4 times with a 30 minute rest period between each shake. The 30 minute intervals allow the water to penetrate the soil aggregates and pull the salts into the solution. Distilled water is used because it has an electrical conductivity of nearly zero. This ensures that any salt we measure comes purely from the soil not from the water itself.



After the shaking intervals were complete, the mixture was a cloudy suspension. To measure the pH and EC accurately, the liquid (extract) had to be separated from the solid soil particles. Funnels were placed directly over clean beakers. Whatman #2 filter paper was folded into a cone shape and seated inside each funnel. The soil water suspension was carefully poured into the filter paper. Then the filtrate or soil extract was collected in the container.

### **THE SATURATED PASTE EXTRACT (SPE) METHOD**

Preparation was done by having 200g of air-dried soil and was placed in a beaker then distilled water was added progressively while stirring until the soil reached a saturated state. Saturation was confirmed when the paste surface glistened, it flowed slightly when tipped at a 45° angle, and it slid cleanly off a spatula. The paste was allowed to stand to reach equilibrium, then transferred to a Buchner funnel where an 80KPa vacuum was applied to collect the filtrate.

Before testing the samples, the BIOBASE 210 Benchtop Meter was calibrated using standard buffer solutions (pH 4.0, 7.0 and 10.0). This ensures the meter identifies what a neutral or a specific salinity level looks like.

The pH and EC electrodes were rinsed with distilled water and gently blotted dry with a lint-free tissue to prevent cross-contamination between samples. The electrodes were submerged into the filtered soil extract and once stabilized, the pH and Electrical Conductivity (EC) readings were taken. The electrodes were rinsed thoroughly with distilled water after every single measurement.

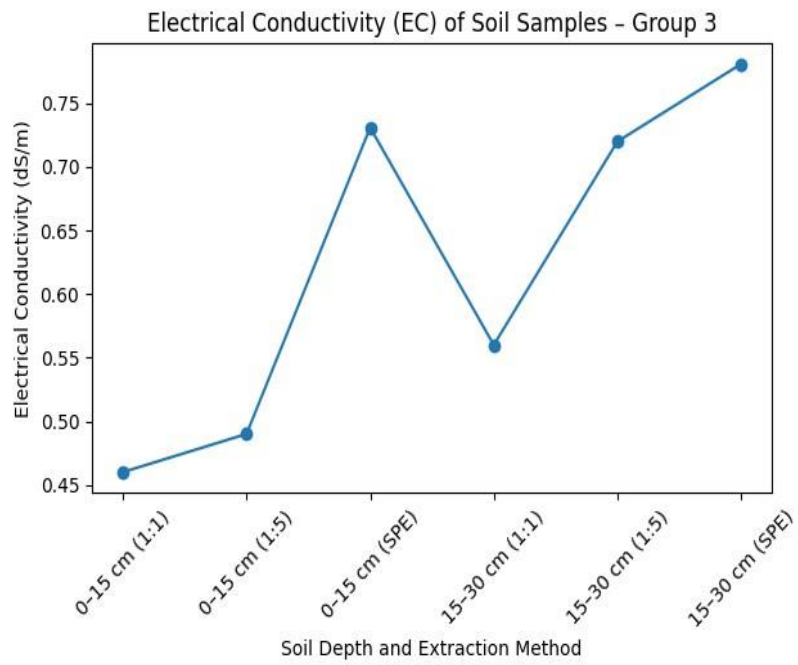


## RESULTS

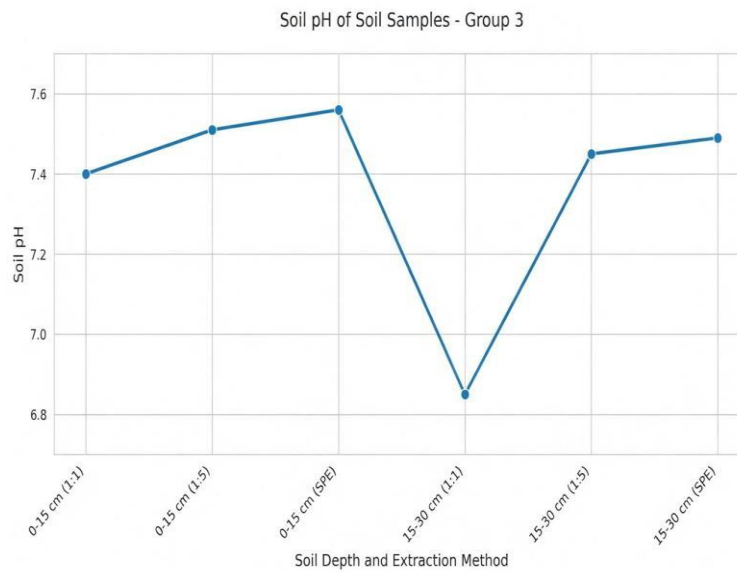
| DEPTHS | RATIO | PARAMETER | GROUP 3 |
|--------|-------|-----------|---------|
| 0-15   | 1:1   | pH        | 7.40    |
|        |       | EC (dS/m) | 0.49    |
|        | 1:5   | pH        | 7.51    |
|        |       | EC (dS/m) | 0.46    |
|        | SPE   | pH        | 7.56    |
|        |       | EC (dS/m) | 0.731   |
| 15-30  | 1:1   | pH        | 6.85    |
|        |       | EC (dS/m) | 0.72    |
|        | 1:5   | pH        | 7.45    |
|        |       | EC (dS/m) | 0.56    |
|        | SPE   | pH        | 7.49    |
|        |       | EC (dS/m) | 0.781   |

The results show that pH values were consistently neutral to slightly alkaline across methods and depths, ranging from 6.85 (1:1 at 15–30 cm) to 7.56 (SPE at 0–15 cm). EC values were low overall (0.46–0.781 dS/m), with SPE consistently yielding the highest readings, followed by 1:1, then 1:5. EC was slightly higher at 15–30 cm than at 0–15 cm in most methods, supporting gradual downward leaching of salts.

## ELECTRICAL CONDUCTIVITY GRAPH



## SOIL PH GRAPH



## **DISCUSSION**

The soil samples collected from depths of 0–15 cm and 15–30 cm showed slight variations in pH and electrical conductivity (EC), which reflects the natural variability of soil properties within a study area. Even within a few meters, soil characteristics can differ due to micro-environmental conditions such as organic matter content, mineral composition and past land management practices.

The pH values obtained ranged from 6.85 to 7.56, indicating that the soil is neutral to slightly alkaline. These conditions are generally favorable for most agricultural crops because essential nutrients remain available within this pH range. Slight differences in pH between the two depths may be influenced by the presence of decomposed organic matter near the surface or the movement of soluble compounds through the soil profile.

Electrical conductivity (EC) values ranged from 0.49 to 0.781 dS/m, which indicates low salinity levels. The slightly higher EC values observed in the saturated paste extract (SPE) method compared to the 1:1 and 1:5 extracts are expected because the SPE method measures salts under more concentrated soil moisture conditions. The results suggest that salts may be moving gradually through the soil profile, possibly due to rainfall or irrigation of water leaching soluble salts downward.

## CONCLUSION

- **General Soil Health**

Salinity: The Electrical Conductivity (EC) values obtained were all below 1.0 dS/m, indicating that the soil is non-saline. This means that there is no immediate risk of osmotic stress or salt toxicity for crops grown in this soil.

- **Vertical Salt Movement (Leaching)**

The comparison between the 0–15 cm and 15–30 cm depths suggest the movement of soluble salts within the soil profile. Slightly higher EC values at the deeper layer may indicate that salts are gradually leaching downward through the soil. This process is beneficial because it prevents the accumulation of salts at the soil surface and indicates adequate drainage conditions.

## RECOMMENDATIONS

- **Continuous Depth Monitoring**

Since the results indicated active leaching (movement of salts from the 0–15 cm layer to the 15–30 cm layer), it is recommended to conduct regular testing of the subsurface profile every two to three seasons. This ensures that salts do not accumulate in the root zone over the long term, which could eventually lead to subsoil salinity.

- **Crop Selection and Management**

The soil's neutral to moderately alkaline pH and low salinity make it highly suitable for a wide range of crops.

- **Irrigation and Drainage Control**

It is vital to maintain good drainage across the field. Avoid excessive irrigation with water of high salt content and effective drainage ensures that any applied salts are flushed below the root zone rather than accumulating at the surface through capillary action and evaporation.

- **Annual Soil Audits**

pH and EC measurements should be conducted annually before the planting season. Consistent monitoring allows for the early detection of chemical shifts, ensuring the soil remains an optimal medium for crop production and sustainable land use.

## REFERENCES

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- **Rhoades, J. D. (1996).** *Salinity: Electrical Conductivity and Total Dissolved Solids*. In: *Methods of Soil Analysis, Part 3: Chemical Methods*. Soil Science Society of America Book Series.
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